



Streaming Joins

Justin Post

Streaming: Joins

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Recap

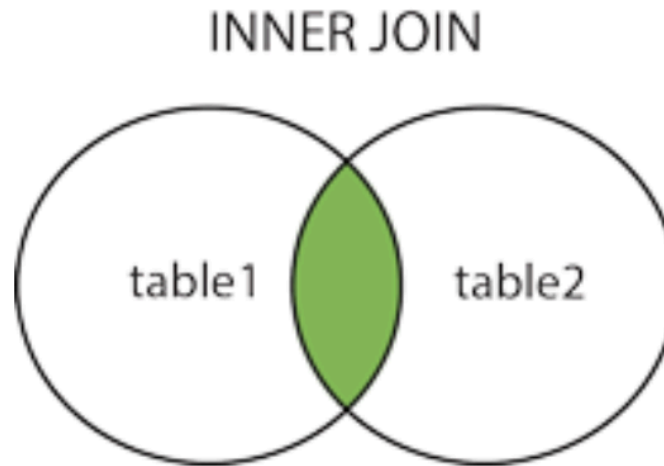
- Create a spark session
1. Read in a stream
 - Stream from a file, terminal, or use something like kafka
 2. Set up transformations/aggregations to do (mostly using SQL type functions)
 - Perhaps over windows
 - Use a watermark to allow for late data
 3. Set up writing of the query to an output source
 - Console (for debugging)
 - File (say .csv)
 - Database
 4. `query.start()` the query!
 - Continues listening until terminated (`query.stop()`)

Can combine two streams or a stream and a static data source

Recall Our Common Joins

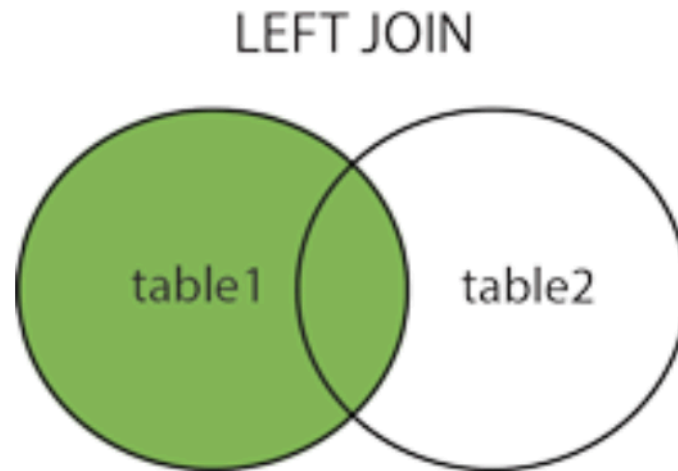
Combining two (or more) tables in SQL is called doing a **join**

- Inner Join: Returns records with matching keys in both tables



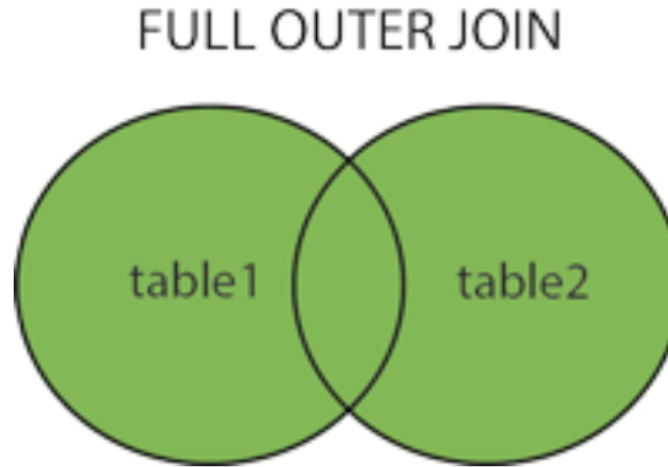
Recall Our Common Joins

- Left (Outer) Join: Returns all records from the 'left' table and any matching records from the 'right' table



Recall Our Common Joins

- (Full) Outer Join: Returns all records when there is a match from the 'left' or 'right' table



Common Joins

- Can do the following **stream to stream** joins in Spark Structured Streaming:
 - Inner
 - Left (must specify watermark on right and time constraints)
 - Right works similarly
 - Full Outer (must specify watermark and time constraints on at least one side)
 - Left Semi (return any rows from the left dataset that were matched with the right table)

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 - Left Semi (return any rows from the left dataset that were matched with the right table)
- Must use **append** output mode
- Cannot do aggregations before joins

Example Inner Join Syntax

- Suppose you have `streamDF1` and `streamDF2`

```
streamDF1.join(streamDF2, "col_id")  # inner join on common column col_id
```

Example Left (Outer) Join Syntax

- Suppose you have `streamDF1` and `streamDF2`
- Each has some watermarks ([code modified from here](#))

```
# Define watermarks
impressionsWithWatermark = impressions \
  .selectExpr("adId AS impressionAdId", "impressionTime") \
  .withWatermark("impressionTime", "10 seconds ") # max 10 seconds late

clicksWithWatermark = clicks \
  .selectExpr("adId AS clickAdId", "clickTime") \
  .withWatermark("clickTime", "20 seconds") # max 20 seconds late
```

Example Left (Outer) Join Syntax

- Suppose you have `streamDF1` and `streamDF2`
- Each has some watermarks ([code modified from here](#))

```
from pyspark.sql.functions import expr

# Left outer join with time range conditions
impressionsWithWatermark.join(
    clicksWithWatermark,
    expr("""
        clickAdId = impressionAdId AND
        clickTime >= impressionTime AND
        clickTime <= impressionTime + interval 1 hour
        """),
    "leftOuter"
)
```

Stream-static Joins

We can also do joins of a stream with a static spark Data Frame

- Suppose streaming DF is on the left and static on the right
 - Inner, left outer, and left semi are supported

Stream-static Joins

We can also do joins of a stream with a static spark Data Frame

- Suppose streaming DF is on the left and static on the right
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Example syntax:

```
streamingDF.join(staticDF, "column", "inner")
```

- For all of these, we then need to write the query!

Example

- Let's jump into pyspark and do a few joins!

Recap

- Can do some stream-to-stream joins and stream-to-static joins
- For stream to stream joins
 - Must use **append** output mode
 - Cannot do aggregations before joins

Course Recap

- 5 V's of Big Data
 - Volume
 - Variety
 - Velocity
 - Veracity (Variability)
 - Value
- Understanding of the Big Data pipeline and basics of handling Big Data
 - Databases/Data Lakes/Data Warehouses/etc.
 - Hadoop
 - Spark
- Modeling data
 - Machine learning algorithms
 - Tuning and testing models
- Common issues seen on data with velocity and Spark Structured Streaming